



Alone with my phone? Examining beliefs about solitude and technology use in adolescence

Robert J. Coplan¹ , Alicia McVarnock¹,
Will E. Hipson² and Julie C. Bowker³

Abstract

In this study, we examined how technology impacts adolescents' perceptions of, and affective responses to solitude, as well as how adolescents' own motivations for solitude (shyness, affinity for aloneness) were related to these reactions. Participants were $N = 437$ adolescents (297 girls; $M_{\text{age}} = 16.15$ years, standard deviation (SD) = .50) who were presented with a series of hypothetical vignettes asking them to imagine themselves in the context of *pure* solitude (alone in their room with the door closed), as well as being physically alone but engaged in increasing levels of virtual social engagement, including passive (e.g., watching videos, scrolling, but no direct social engagement), active (e.g., texting), and audio-visual (e.g., Facetime) technology use. Following each vignette, participants reported their perceptions of being alone and positive/negative affective responses. We also measured general motivations for solitude (shyness, affinity for aloneness). Among the results, adolescents perceived themselves as less alone in vignettes depicting increasing virtual social engagement. Affective benefits of increased virtual engagement were also found (e.g., less loneliness/boredom/sadness, greater social connection/contentment). However, these effects were moderated by solitude motivations, with different patterns evident as a function of participant shyness and affinity for aloneness. Findings highlight the importance of considering the nature of adolescents' technology use when alone, as well as motivations for solitude, when considering links between solitude and well-being.

Keywords

Adolescence, solitude, technology, emotion

Adolescence is a developmental period marked by the emergence of solitude as an important context for positive development (Bowker et al., 2016), as well as the near ubiquitous use of technology (Vogels, 2019). Moreover, contemporary computer-mediated communication technologies (e.g., smartphones) now make it possible for adolescents to be physically alone yet virtually engaging with countless others. Yet, these two critical contributors to adolescent development continue to be studied predominantly separately, and as a result, little is currently known about adolescents' perceptions and experiences of solitude and technology use. In the present study, we considered the intersection of these two phenomena by examining adolescents' perceptions of, and affective responses to, solitude when physically alone but under different conditions of technology use. Of interest were adolescents' perceptions of being alone but under different levels of *virtual engagement* (e.g., watching videos, texting, interacting audio-visually). We also examined whether adolescents' motivations for solitude (i.e., affinity for aloneness, shyness) were related to their perceptions and anticipated affective responses to *pure* solitude (physically alone without using technology to interact with others; Nguyen et al., 2018) and the various conditions of being physically alone but engaging in different technologies.

Experiences of Solitude in Adolescence

Developmental theorists and researchers have proposed several reasons why adolescence is an important and unique period for understanding experiences of solitude (Bowker et al., 2016). For example, time spent alone increases from early to late childhood and into adolescence, along with growing autonomy in terms of how to spend leisure time (Coplan et al., 2019). This has important implications for how adolescents experience solitude, as time alone engaged in *intrinsically motivated* activities is more likely to be associated with positive affective states (Coplan et al., 2021). Adolescents also obtain privacy through solitude

¹ Carleton University, Canada

² Global Spatial Technology Solutions, Canada

³ University at Buffalo, The State University of New York, USA

Corresponding author:

Robert J. Coplan, Carleton University, 1125 Colonel By Drive, Ottawa, ON K1S 5B6, Canada.

Email: robert.coplan@carleton.ca

(an increasingly valued experience at this age) and seek relief from external social pressures and self-monitoring (Keijsers et al., 2010). Finally, time alone may facilitate self-exploration and identity development, which are central developmental tasks during adolescence (Goossens, 2014).

Accordingly, results from several recent studies support the potential benefits of solitude in adolescence (Borg & Willoughby, 2022; Coplan et al., 2021; Thomas & Azmitia, 2019). As compared to later childhood, general attitudes about solitude become increasingly positive during adolescence (Danneel et al., 2018), and by around age 16, adolescents view the desire to spend time alone as highly normative (Wood et al., 2021). Moreover, among adolescents who spend the most time alone, the majority: (1) appear to do so out of preference/enjoyment (rather than anxiety/avoidance); (2) engage in intrinsically motivated solitary activities (e.g., hobbies, listening to music); and (3) report comparable levels of trait positive/negative affect to their more sociable counterparts (Coplan et al., 2021; Hipson et al., 2021).

Solitude and Technology

Larson (1990) argued that individuals who are physically alone while watching television or listening to music are still spending time in solitude, as these technologies do not directly command reciprocal interactions or require feedback. However, talking on the phone while physically alone (i.e., an interactive technology) would not. Applying this argument in the context of contemporary technology, *passive* technology use (e.g., watching videos or scrolling social media newsfeed) while alone would be considered solitude, whereas *active* technology use (e.g., interactions with others via text, audio, or video) would not. Interestingly, many adolescents do not entirely agree with Larson's (1990) conceptualization of solitude. Hipson et al. (2021) asked adolescents to list the three things they did most while alone during the last week and 87% mentioned using technology. Consistent with Larson's (1990) conceptualization, passive screen time (e.g., Netflix) was mentioned most often (41%). However, respondents also recalled engaging in more socially interactive forms of technology while "alone," including texting, talking on the phone, and even audio-visual communication (e.g., FaceTime). These findings indicate a potential gap between how solitude is conceptualized by researchers/theorists versus how solitude is understood in practice by adolescents. Accordingly, one goal of this study was to examine to what degree adolescents consider themselves to be "alone" when physically separated from others and under contexts of increasing virtual engagement.

Screen time has been increasing year by year, especially among adolescents and young adults (Vogels, 2019). Still, there remains considerable debate regarding the costs and benefits of technology use for adolescents (Hollis et al., 2020). However, it has become clear that both how (i.e., type of use)—and under what circumstances (i.e., social context)—young people engage with technology should be considered when understanding the potential impact of screen time on adolescents' well-being (Burnell et al., 2021). It is important to better understand these issues, because although many adolescents and young adults use time alone to engage with technology (Thulin & Vilhelmson, 2019), technology use while alone may also interfere and disrupt the positive aspects of these solitary experiences (Kushlev et al., 2016), particularly for those who value and enjoy solitude (Diefenbach & Borrmann, 2019; Thomas et al., 2021).

The Present Study

The overall aim of this study was to examine adolescents' perceptions and affective responses to physical solitude while under different conditions of technology use. We operationalized physical solitude as being alone in one's bedroom with the door closed. This is a common context for solitude among adolescents and provides a context for privacy seeking and identity development (Reid, 2012). Participants were first asked to imagine themselves in *pure* solitude (Nguyen et al., 2018) and then as still physically alone but in contexts of increasing virtual engagement with others.

The progression of virtual engagement contexts was based upon theories characterizing how technology is used to facilitate social interaction. For example, according to *Media Synchronicity Theory* (Dennis et al., 2008), social communication effectiveness is influenced by the quality and transmission of cues, including immediacy of feedback, ability to transfer nonverbal and verbal cues, and degree of personalization. Similarly, according to *Social Presence Theory* (Kock, 2004), increased media richness amplifies social presence, the awareness of the physical presence or closeness of another during an interaction. Drawing upon these theoretical perspectives, we predicted that despite being in physical solitude in all circumstances, adolescents would perceive themselves as progressively less alone with increasing levels of virtual engagement.

Another study aim was to explore adolescents' affective reactions to imagining themselves as being alone while engaging differentially with technology. Here, we considered both general positive (i.e., content, socially connected) and negative (i.e., bored, sad, lonely) responses. In general, experiences of pure solitude tend to evoke negative emotions among adolescents and young adults (Wilson et al., 2014), particularly as compared to being alone and engaging in other activities (Hipson et al., 2021). Of particular relevance to the present study, Thomas et al. (2021) also recently reported that college students generally reported better mood (e.g., less boredom, less loneliness) when alone on their devices than when in pure solitude. Accordingly, we expected adolescents to report more boredom, sadness, and loneliness, as well as less contentment and social connectedness when imagining themselves participating in solitude contexts with progressively less virtual engagement.

Finally, we explored the role of adolescents' own motivations for solitude. For example, adolescents with higher *affinity for aloneness* (Goossens, 2014) have more positive attitudes toward solitude (Wood et al., 2021) and tend to enjoy spending time alone (Daly & Willoughby, 2020). In this regard, affinity for aloneness was expected to predict more generally positive responses to solitude. Results concerning the link between affinity for aloneness and overall frequency of smartphone use have been somewhat mixed, with some studies reporting a negative relation (Diefenbach & Borrmann, 2019) and others not direct association (Birmingham et al., 2021). However, adolescents with higher *affinity for aloneness* do appear to be more inclined to spend time alone without engaging in active technology use (Diefenbach & Borrmann, 2019; Thomas et al., 2021). In this regard, the interactive nature of technology use of each condition was expected to impact the link between affinity for aloneness and positive reactions to solitude, such that this association would be stronger in the pure solitude and passive technology conditions, but increasingly attenuated in more socially interactive contexts (e.g., active, audio-visual).

In contrast, *shy* adolescents tend to retreat into solitude due to socio-evaluative concerns, despite a desire for social interaction,

(Bowker et al., 2016). Shy adolescents also report more negative cognitions (e.g., rumination) and emotions (e.g., worry) when alone (Blöte et al., 2019; Coplan et al., 2021). However, audio-visual communications involving synchronous interactions and visual cues may also evoke socio-evaluative concerns (Chuk, 2018). Passive/active technology use may provide a safer and less stressful “middle ground” whereby shy adolescents may (at least partially) satisfy their desire for social interaction (Burnell et al., 2021). In this regard, shyness was expected to predict more negative reactions in both the pure solitude and audio-visual communication conditions.

Finally, in terms of potential gender differences, there is some evidence to suggest that adolescent boys spend more time physically alone than adolescent girls (Hipson et al., 2021). In addition, it has been suggested that shyness and affinity for aloneness may be viewed more negatively as characteristics among males than females because social withdrawal violates male gender stereotypes pertaining to dominance and assertion (Doey et al., 2014). There are also some reported gender differences in technology use (e.g., girls spend more time on social media and texting, boys spend more time gaming) and at least some evidence to suggest that extensive technology use can have stronger negative mental health implications for girls than boys (Keles et al., 2020; Twenge & Martin, 2020). In this regard, we thought it was relevant to include gender as a variable in our main analyses. However, because we did not have an explicit rationale pertaining to the specific role of gender in adolescents’ reactions to solitude and technology, these analyses should be considered exploratory in nature.

Method

Participants

Participants were $N=437$ adolescents (140 boys, 297 girls) ages 15–18 years ($M=16.15$, standard deviation (SD)=.50) attending public high schools in Eastern Ontario, Canada. Schoolboards did not permit collecting individual data on ethnicity or socioeconomic status. However, participating schools were from a mix of urban, suburban, and rural neighborhoods. According to the most recent publicly available data, about 85% of students in these schools had a parent who completed post-secondary education and 45% self-identified with a racial background other than White (OCDSB, 2012).

Procedure

Following approval from the *Carleton University Research Ethics Board* (Project #107297) and the *School Board (Ottawa-Carleton Research and Evaluation Advisory Committee, approved 3 October 2018)*, written parental consent (consent rate=70%) and online assent from adolescent participants (assent rate=98%) were obtained. Participants were provided with a link to the online study, which they completed during class-time on a laptop or Smartphone. Data collection was anonymous and took place between October 2018 and May 2019.

Measures

For the purposes of this study, we created a series of hypothetical vignettes depicting adolescents in physical solitude while

under different conditions of technology use,¹ adapted from previous research assessing attitudes and beliefs about social and non-social behaviors in different contexts (e.g., Coplan et al., 2007). Characterizations of the different types of technology described in the vignettes were drawn from studies on active and passive technology use (Verduyn et al., 2015). Adolescents were first asked to imagine themselves in pure solitude (“alone in their room with the door closed”), and then under such conditions but with increasing virtual engagement: passive technology (e.g., watching videos, scrolling social media newsfeed, but no direct social engagement); active technology (e.g., texting); and audio-visual communication (e.g., Facetime). Following each vignette, participants indicated their positive (i.e., content, socially connected) and negative (i.e., bored, sad, lonely) emotional responses (on a 5-point from “*Not at all*” to “*Extremely*”). As well, for all but the pure solitude vignette, participants also indicated whether they perceived themselves as “alone” (on a 5-point scale from “*Strongly Disagree*” to “*Strongly Agree*”).

Finally, two motivations for social withdrawal were assessed: (1) shyness (*Revised Cheek & Buss Shyness Scale*, Cheek & Buss, 1981; 20 items, $\alpha=.75$; e.g., “I feel inhibited in social situations”) and (2) affinity for aloneness (*Loneliness and Aloneness Scale for Children and Adolescents*, Marcoen et al., 1987; 12 items, $\alpha=.85$; e.g., “When I am alone, I quiet down”). Both scales have excellent psychometric properties and have been previously used with adolescents (Coplan et al., 2021; Danneel et al., 2018).

Results

Analytic Plan

The percentage of missing data ranged from 1% (gender) to 6% (affinity for aloneness). R (v. 4.1.0) was used for all analyses using the lme4 and lmerTest packages. Vignette was treated an *ordinal* variable, representing increasing levels of virtual engagement (i.e., alone, passive, active, and audio-visual technology use). To examine main effects, linear models were first constructed separately for each outcome (Being Alone, Loneliness, Social Connectedness, Boredom, Sadness, and Contentment). Gender, Vignette, Shyness, and Affinity for Aloneness were treated as fixed effects. Because each participant responded to each vignette, participants were treated as random effects. Next, the following interaction terms were added to each model: (1) Vignette $\times$ Shyness; (2) Vignette $\times$ Affinity for Aloneness; and (3) Vignette $\times$ Gender. Complete results are displayed in Table 1. Although Vignette conditions are ordered, each condition represents a distinct form of virtual engagement. Moreover, for interaction effects, it was of particular theoretical interest to examine each discrete condition. As such, for interaction effects involving covariates, simple effects were probed with regression analyses separately for each Vignette (see Table 2 for relevant means).

Primary Analyses

Main Effects. Results indicated significant main effects of Vignette for perceptions of being alone, loneliness, boredom, sadness, social connectedness, and contentment. Overall, increased virtual engagement across vignette conditions was

Table 1. Random Intercept Models of Effect of Technology Use, Gender, Shyness, and Affinity for Aloneness on Perceptions of and Affective Responses to Being Alone.

Effect	Lonely (n = 1609)				Socially connected (n = 1603)				Bored (n = 1606)			
	b	Partial η^2	95% CI		b	Partial η^2	95% CI		b	Partial η^2	95% CI	
			LL	UL			LL	UL			LL	UL
Vignette	-.31***	.23	-.34	-.27	.65***	.45	.30	.68	-.40***	.25	-.44	-.37
Gender	.01	.00	-.13	.15	-.09	.00	-.22	.04	-.07	.00	-.22	.08
Shyness	.31***	.05	.17	.45	-.03	.00	-.16	.11	.06	.00	-.08	.20
Affinity	-.28***	.04	-.42	-.13	-.09	.00	-.22	.05	-.33***	.05	-.48	-.18
Vignette $\times$ Gender	-.04	.00	-.11	.02	.02	.00	-.07	.10	-.04	.00	-.12	.05
Vignette $\times$ Shyness	-.20***	.03	-.27	-.14	.15***	.01	.06	.23	-.04	.00	-.12	.04
Vignette $\times$ Affinity	.29***	.06	.23	.36	-.06	.00	-.15	.02	.33***	.05	.25	.41

Effect	Content (n = 1613)				Sad (n = 1606)				Alone (n = 1198)			
	b	Partial η^2	95% CI		b	Partial η^2	95% CI		b	Partial η^2	95% CI	
			LL	UL			LL	UL			LL	UL
Vignette	.26***	.14	.22	.29	-.26***	.21	-.29	-.23	-.86***	.50	-.92	-.80
Gender	.01	.00	-.13	.15	.09	.00	-.08	.26	-.16	.00	-.34	.02
Shyness	-.01	.00	-.15	.12	.35***	.04	.19	.51	.01	.00	-.17	.19
Affinity	.23**	.02	.09	.36	.00	.00	-.16	.17	.02	.00	-.16	.20
Vignette $\times$ Gender	.13***	.00	.06	.21	-.09**	.00	-.16	-.03	-.00	.00	-.14	.13
Vignette $\times$ Shyness	.05	.00	-.02	.13	-.17***	.03	-.23	-.11	-.25***	.02	-.37	-.12
Vignette $\times$ Affinity	-.23***	.03	-.31	-.16	.14***	.02	.08	.20	.14*	.00	.01	.27

Note. 95% CI = 95% confidence interval of regression coefficient; LL = lower limit. UL = upper limit.

* $p < .05$. ** $p < .01$. *** $p < .001$.

Table 2. Means (and Standard Deviations) for Affect by Technology Use Condition.

Response	Technology use condition			
	Alone	Passive	Active	Audio-visual
Alone	—	3.46 (1.20) n = 419	2.43 (1.15) n = 413	1.75 (1.00) n = 404
Lonely	2.11 (1.13) n = 421	3.03 (1.10) n = 420	1.57 (.84) n = 416	1.23 (.61) n = 412
Socially connected	1.98 (1.04) n = 421	3.02 (.06) n = 421	3.61 (.95) n = 413	3.94 (.98) n = 410
Bored	2.64 (1.27) n = 420	2.29 (1.15) n = 421	1.86 (.93) n = 415	1.46 (.78) n = 412
Content	3.06 (1.05) n = 420	2.96 (.96) n = 423	3.35 (.88) n = 418	3.77 (.94) n = 413
Sad	2.11 (1.19) n = 421	1.93 (1.10) n = 419	1.61 (.89) n = 416	1.35 (.72) n = 412

Note. Responses measured on a 5-point scale.

negatively associated with perceptions of aloneness ($b = -0.86$, standard error (SE) = 0.03, $t = -28.01$, $p < .001$), loneliness ($b = -.31$, $SE = .02$, $t = -18.90$, $p < .001$), boredom ($b = -.40$, $SE = .02$, $t = -20.29$, $p < .001$), and sadness ($b = -.26$, $SE = .01$, $t = -17.83$, $p < .001$). Increased virtual engagement was also positively associated with social connectedness ($b = .65$, $SE = .02$, $t = 31.35$, $p < .001$) and contentment ($b = .26$, $SE = .02$, $t = 13.92$, $p < .001$). There were no significant main effects of Gender.

There were significant main effects of (participant) Shyness for loneliness and sadness. Overall, Shyness was positively associated with loneliness ($b = .31$, $SE = .07$, $t = 4.39$, $p < .001$) and sadness ($b = .35$, $SE = .08$, $t = 4.28$, $p < .001$). In addition, there were significant main effects of (participant) Affinity for Aloneness for loneliness, boredom, and contentment. Overall, Affinity for Aloneness was negatively associated with loneliness ($b = -.28$, $SE = .07$, $t = -3.81$, $p < .001$) and boredom ($b = -.33$,

$SE = .07$, $t = -4.43$, $p < .001$), and positively associated with contentment ($b = .23$, $SE = .07$, $t = 3.19$, $p = .002$).

Interactions

Vignette $\times$ Gender. Significant interactions were found for Vignette $\times$ Gender for contentment and sadness. In the pure solitude, passive, and active vignettes, there were no significant gender differences for contentment (pure solitude: $b = -.14$, $SE = .11$, $t = -1.25$, $p = .212$; passive: $b = -.14$, $SE = .10$, $t = -1.32$, $p = .187$; active: $b = .11$, $SE = .10$, $t = 1.16$, $p = .247$). In the audio-visual vignette, however, girls reported greater contentment than boys ($b = .24$, $SE = .10$, $t = 2.31$, $p = .021$).

Although gender was not significantly associated with sadness across vignettes (pure solitude: $b = .22$, $SE = .12$, $t = 1.77$, $p = .077$; passive: $b = .16$, $SE = .12$, $t = 1.40$, $p = .162$; active: $b = .00$, $SE = .10$, $t = .00$, $p = .999$; audio-visual: $b = -.02$, $SE = .08$, $t = -.21$, $p = .831$), given the significant interaction, it can be inferred that the pattern of gender differences in the pure solitude vignette (i.e., girls > boys) became attenuated with increasing levels of interactive technology use.

Vignette $\times$ Shyness. Results revealed significant Vignette $\times$ Shyness interactions for perceptions of being alone, loneliness, social connectedness, and sadness. In the pure solitude vignette, Shyness was positively associated with loneliness ($b = .62$, $SE = .11$, $t = 5.64$, $p < .001$) and negatively associated with social connectedness ($b = -.31$, $SE = .11$, $t = -2.89$, $p = .004$). In the passive vignette, Shyness was positively associated with loneliness ($b = .47$, $SE = .11$, $t = 4.31$, $p < .001$) and perceptions of being alone ($b = .36$, $SE = .13$, $t = 2.84$, $p = .005$); however, Shyness was not significantly associated with feelings of social connectedness ($b = -.11$, $SE = .11$, $t = -.99$, $p = .322$).

In the active and audio-visual vignettes, Shyness was not significantly associated with perceptions of being alone (active: $b = -.19$, $SE = .12$, $t = -1.54$, $p = .123$; audio-visual: $b = -.15$, $SE = .10$, $t = -1.43$, $p = .154$) or loneliness (active: $b = .05$, $SE = .09$, $t = .54$, $p = .589$; audio-visual: $b = .08$, $SE = .06$, $t = 1.47$, $p = .141$). In the active vignette, but not the audio-visual vignette, Shyness was positively associated with feelings of social connectedness (active: $b = .26$, $SE = .10$, $t = 2.62$, $p = .009$; audio-visual: $b = .05$, $SE = .10$, $t = .53$, $p = .599$). Shyness was positively associated with sadness across vignettes; however, the magnitude of the associations decreased with increased virtual engagement (pure solitude: $b = .68$, $SE = .12$, $t = 5.65$, $p < .001$; passive: $b = .35$, $SE = .11$, $t = 3.17$, $p = .002$; active: $b = .22$, $SE = .09$, $t = 2.37$, $p = .018$; audio-visual: $b = .16$, $SE = .07$, $t = 2.13$, $p = .034$).

Vignette $\times$ Affinity for Aloneness. There were significant Vignette $\times$ Affinity for Aloneness interactions on perceptions of being alone, loneliness, boredom, contentment, and sadness. In the pure solitude and passive vignettes, Affinity for Aloneness was negatively associated with loneliness and boredom (pure-solitude-lonely: $b = -.76$, $SE = .11$, $t = -6.74$, $p < .001$; pure-solitude-bored: $b = -.90$, $SE = .13$, $t = -6.99$,

$p < .001$; passive-lonely: $b = -.36$, $SE = .11$, $t = -3.25$, $p = .001$; passive-bored: $b = -.43$, $SE = .12$, $t = -3.49$, $p < .001$), and positively associated with contentment (pure solitude: $b = .69$, $SE = .11$, $t = 6.34$, $p < .001$; passive: $b = .22$, $SE = .10$, $t = 2.18$, $p = .030$). Affinity for Aloneness was also negatively related to sadness in the pure solitude vignette, but not the passive vignette (pure-solitude-sad: $b = -.29$, $SE = .12$, $t = -2.37$, $p = .019$; passive-sad: $b = .02$, $SE = .12$, $t = .16$, $p = .870$).

In the active and audio-visual technology vignettes, Affinity for Aloneness was not associated with loneliness (active: $b = .05$, $SE = .09$, $t = .60$, $p = .546$; audio-visual: $b = .02$, $SE = .06$, $t = .39$, $p = .697$), boredom (active: $b = -.08$, $SE = .10$, $t = -.78$, $p = .434$; audio-visual: $b = .08$, $SE = .08$, $t = .97$, $p = .334$), sadness (active: $b = .15$, $SE = .09$, $t = 1.60$, $p = .110$; audio-visual: $b = .13$, $SE = .08$, $t = 1.72$, $p = .087$), or contentment (active: $b = .02$, $SE = .10$, $t = .20$, $p = .845$; audio-visual: $b = -.02$, $SE = .10$, $t = -.19$, $p = .848$). Affinity for Aloneness was not significantly associated with perceptions of being alone across all vignettes (passive: $b = -.11$, $SE = .13$, $t = -.89$, $p = .372$; active: $b = .00$, $SE = .12$, $t = .00$, $p = .998$; audio-visual: $b = .19$, $SE = .11$, $t = 1.75$, $p = .081$), but given the significant interaction, it can be inferred that the negative association with perceptions of being alone in the passive vignette differed from the positive association with being alone in the audio-visual vignette.

Discussion

The goal of this study was to examine how virtual engagement impacts adolescents' perceptions of, and anticipated affective responses to, solitude. We also explored how adolescents' own motivations for solitude (i.e., shyness, affinity for aloneness) were related to these reactions. Overall, adolescents perceived themselves as less alone in vignettes depicting physical solitary experiences with increasing virtual social engagement. Affective benefits of increased virtual engagement were also found (i.e., less loneliness, sadness, and boredom, greater social connectedness and contentment). However, these effects were moderated by motivations for solitude, with different patterns of results for affinity for aloneness and shyness. Findings highlight the importance of considering both internal (motivational) and contextual (technology use) factors when considering experiences of solitude among adolescents.

Solitude and Technology Use

Solitude has typically been examined dichotomously (i.e., physically "alone" vs. "with others"; see Coplan et al., 2021). However, as hypothesized, results from the present study suggest that adolescents perceive solitude as a *gradient* when using technology, with increased virtual social engagement being negatively associated with feelings of being alone. In this regard, it seems that researchers should go beyond the simple consideration of solitude as a *physical* separation from others (Wilson et al., 2014) to explicitly assess (and account for) nuances in solitary experiences today, which are due to new technologies. In this regard, Campbell and Ross (2022) recently argued that solitude should be reconceptualized for the digital era to take into account for face-to-face and computer-mediated forms of communication. In this regard, the authors suggest that notion of physical solitude

should be replaced by *social aloneness*, which can be viewed as a continuum (referred to as *shades* of solitude) that takes into account expectations for computer-mediated communication and aspects of accessibility to others.

Given the near ubiquitous use of technology in adolescence (Vogels, 2019), including while physically alone (Hipson et al., 2021; Thulin & Vilhelmson, 2019), it has become increasingly clear that simply asking adolescents how much time they spend alone may no longer adequately capture their experiences of solitude. Our results are novel as they suggest that not only do adolescents consider the role of technology when evaluating to what degree they consider themselves to be alone, but that technology may also impact their experiences in solitude.

Consistent with our hypotheses, adolescents in the current study anticipated feeling more bored, sad, and lonely as hypothetical virtual engagement decreased. They also anticipated feeling less content and socially connected. Overall, previous studies have suggested that time spent alone is associated with less positive and more negative affect compared to that spent with others (e.g., Matias et al., 2011). This is particularly true for episodes of *pure* solitude, where participants are alone in a room without engaging with technology (Nguyen et al., 2018) or other activities (Wilson et al., 2014). Most recently, Thomas et al. (2021) reported that adult participants reported being consistently in a better mood while on their devices alone as compared to when they were in pure solitude. Our findings are also consistent with theoretical perspectives outlining how different types of technologies might serve to facilitate virtual social interaction. For example, *Media Synchronicity Theory* (Dennis et al., 2008) and *Social Presence Theory* (Kock, 2004) posit that increased media richness will heighten the effectiveness of social communication and amplify social presence. Moreover, given that humans generally report more positive emotions when interacting with others than while alone (Matias et al., 2011), our results suggest that increased virtual engagement may also offer some of this affective boost to adolescents.

Role of Motivations for Solitude

A secondary goal of this study was to explore how adolescents' motivations for solitude influence their experience of the intersection between solitude and technology. In line with our expectations, findings indicate that both affinity for aloneness and shyness *uniquely* contribute to how adolescents perceive and anticipate responding to solitude in different contexts of technology use. Findings are described in detail below.

Affinity for Aloneness

Results regarding affinity for aloneness generally supported hypotheses. For example, as expected, *affinity for aloneness* was only associated with more positive anticipated experiences of being alone (i.e., greater contentment, less boredom/sadness/loneliness) in the pure solitude and passive technology vignettes. In contrast, in the active and audio-visual vignettes, these positive experiences of solitude were attenuated. These results are consistent with previous research suggesting that adolescents higher in affinity for aloneness particularly enjoy spending time in pure solitude without engaging in active technology use (Coplan et al., 2021; Thomas et al., 2021). Moreover, adolescents

high in affinity for aloneness may perceive increased virtual engagement as intrusive, thereby attenuating their beliefs that time alone would be experienced more positively in the active and audio-visual vignettes.

It should be noted that affinity for aloneness was not linked to social connectedness across vignettes. Although these findings were not entirely in line with the hypothesis that affinity for aloneness would predict positive affective experiences under contexts of pure solitude and passive technology use, findings regarding social connectedness make some intuitive sense, as episodes of pure solitude and passive solitary activities (e.g., watching Netflix) are not inherently social.

Shyness

Compared to affinity for aloneness, *shyness* was associated with a very different set of responses. Results from the present study provided partial support for our hypotheses that shyness would relate to negative affect under conditions of pure solitude and audio-visual technology use, but positive affect when engaging in technology use passively and actively. For example, as expected, shyness was associated with a generally negative anticipated experience of being alone (i.e., more sadness and loneliness, less social connectedness) in the pure solitude vignette. This finding is in line with the idea that adolescents high in shyness want to engage with others (Bowker et al., 2016). It also supports previous studies suggesting that shy adolescents are particularly prone to spending time alone worrying and ruminating (Blöte et al., 2019; Coplan et al., 2021). At the highest level of virtual interaction (i.e., audio-visual communication), shyness was surprisingly no longer associated with loneliness or social connectedness. Shyness was, however, related to greater anticipated sadness (albeit at a lesser magnitude than in the pure solitude vignette), suggesting that shy adolescents may experience some negative emotions in contexts with synchronous interactions and visual cues. Taken together, although shyness may represent less of a detriment in audio-visual settings, shy adolescents' desires to socially engage may be hindered, at least somewhat, by socio-evaluative concerns in situations perceived as socially stressful (Bowker et al., 2016).

As expected, shyness was associated with more positive experiences of solitude in the active technology use vignette. For example, shyness did not predict anticipated loneliness. Shyness was also significantly *positively* associated with social connectedness. Interacting with friends through text may allow shy adolescents to respond on their own time and alleviate concerns about visible cues of embarrassment or anxiety (e.g., blushing). This could help to diminish the perceived stress associated with higher pressure social situations wherein interaction is synchronous and visible and in turn facilitate feelings of connection to others. At the same time, it may allow shy adolescents to escape some of the negative feelings associated with solitude, suggesting that there may indeed be a technological *sweet spot* for shy adolescents in terms of virtual engagement. However, it should be noted that shyness was not associated with contentment or boredom across vignettes.

Unfortunately, shy individuals may not be taking advantage of the potential benefits afforded to them by passive computer-mediated communications—results from a recent meta-analysis indicated that shyness was negatively associated with active

social media use (Appel & Gnambs, 2019). Moreover, socially anxious individuals tend not to find the support they are seeking online, which may further exacerbate their negative responses to this context (O'Day & Heimberg, 2021).

Conclusion, Limitations, and Future Directions

Overall, our findings suggest that experiences of solitude in adolescence are varied, complex, and influenced by a range of individual and technology-related contextual factors. It has been postulated that technology use can disrupt adolescents' positive experiences of solitude (Kushlev et al., 2016). However, our findings suggest that adolescents' penchant for virtual engagement may not necessarily be a bad thing. Adolescents generally appear to perceive solitude as more positive and less negative as virtual engagement increases. Among shy adolescents, pure solitude and passive technology use while alone may interfere with positive affect, whereas engaging in active (but not audio-visual) technology use may confer some benefits. Importantly, affinity for aloneness was the only variable not accompanied by negative affective experiences of solitude across vignettes. Findings support our hypothesis that adolescents with higher affinity for aloneness would have more positive attitudes toward solitude. Taken together, it appears that what it means to be alone and how solitude is experienced in adolescence are intimately linked with virtual engagement and motivations for solitude. Notwithstanding, results are preliminary and should be considered with some limitations in mind, with an eye toward future research, which we describe next.

First, with the hypothetical vignettes, we evaluated adolescents' beliefs about how they would feel in solitude under different conditions of technology use. In the pure solitude vignette, adolescents' perceptions of being alone were not assessed, making it important for future studies to examine how perceptions of being alone in pure solitude compare to those while using technology. Moreover, although the pure solitude vignette is proposed as a context of physical solitude without using technology to interact with others, participants were not specifically directed to imagine themselves alone in their bedrooms *without technology*. As such, the possibility of adolescents imagining themselves using technology in the pure solitude condition cannot be ruled out. Future research using experimental manipulation or experience sampling is also required to assess the degree to which adolescents' anticipated responses to these circumstances align with their emotions in the moment—and how they compare to the context of face-to-face (in-person) interactions. Such studies will also help clarify how solitude is defined in practice by adolescents, because although participants in the present study reported perceiving themselves as *less* alone as they engaged more actively with others through technology, we cannot conclude that they did not feel “alone” (Hipson et al., 2021).

In addition, literature on *social snacking* suggests that different forms of passive technology use might have alternative implications for adolescents (Gardner et al., 2005). In particular, when adolescents spend increased time passively scrolling or viewing others' social media profiles to meet social needs (i.e., social snacking), they may feel lonelier and less socially connected because their desire for genuine connection goes unrealized (Clark et al., 2018; Gardner et al., 2005). Watching television or

videos need not be socially driven. To gain a better understanding regarding how passive technology use impacts adolescents' perceptions of and affective responses to solitude, future studies should distinguish between different passive technology use behaviors. Additional types of technology use that may contribute uniquely to experiences of solitude (e.g., *gaming*) may also be explored.

Another limitation pertains to possible context effects. First, vignettes were presented to all participants in order of increasing virtual engagement. Future studies should randomize the order in which vignettes are presented to account for possible order effects. Since questionnaires were completed during class in the presence of others, it may also be useful to examine whether findings hold in solitary testing environments. Finally, while most adolescents have their own bedroom (Currie et al., 1997), some share bedrooms with siblings. Having a private bedroom may impact adolescents' experiences of solitude and psychological well-being (see Moroney, 2019). As participants were asked to imagine themselves alone in their bedroom for each vignette, it is possible that participants with shared bedrooms responded differently than their peers with their own bedrooms. As such, whether adolescents have their own private bedrooms should be considered in future research.

It should also be noted that although the construct of *affinity for aloneness* is generally conceptualized as representing a non-fearful preference for solitude (Goossens, 2014), the measure also includes items pertaining to seeking solitude to complete specific tasks (e.g., “I want to be alone to do some things”) and to avoid social interaction (e.g., “retire from others because they disturb me with their noise”). More recently developed measures have focused more specifically on the role of *self-determination* in adolescents' motivations seeking solitude (Thomas & Azmitia, 2019). From this perspective, self-determined motivations (e.g., enjoying solitude for its own sake or for the benefits it brings), which share similarities with affinity for aloneness, are contrasted with not-self-determined motivations (e.g., avoiding others or about feeling that one “should” be in solitude), which share similarities with shyness. A more detailed discussion of these somewhat nuanced differences is beyond the scope of this study (but see Coplan et al., 2019), but future research should consider replicating and extending the present findings using such alternative assessments.

As well, although not a central focus on the current study, some gender differences were found. For example, females anticipated being more content than their male counterparts while interacting audio-visually, and more sad in the alone condition. Still, females accounted for almost 70% of the present sample. Future research should explore these gender differences further using more balanced samples. Whether current findings generalize across different developmental periods should also be examined (Coplan et al., 2019). Finally, data were collected before the onset of the COVID-19 global pandemic. Future studies should evaluate how the nature of solitary experiences, and how adolescents are impacted by them, have changed since then. In addition, our findings suggest that using technology to engage virtually with others could potentially improve adolescents' experiences of social isolation, which may have important implications given that many young people remain separated from friends and family under the implementation of social distancing and lockdowns.

Funding

The author(s) disclosed receipt of the following financial support for the research, authorship, and/or publication of this article: This research was support by a Social Science and Humanities Research Council of Canada Insight Grant (435-2017-0849) to authors R.J.C. and J.C.B.

ORCID iDs

Robert J. Coplan  <https://orcid.org/0000-0003-3696-2108>

Will E. Hipson  <https://orcid.org/0000-0002-3931-2189>

Julie C. Bowker  <https://orcid.org/0000-0001-7055-9158>

Note

1. Available from the authors, upon request

References

- Appel, M., & Gnambs, T. (2019). Shyness and social media use: A meta-analytic summary of moderating and mediating effects. *Computers in Human Behavior*, 98, 294–301. <https://doi.org/10.1016/j.chb.2019.04.018>
- Bermingham, L., Meehan, K. B., Wong, P. S., & Trub, L. (2021). Attachment anxiety and solitude in the age of smartphones. *Psychoanalytic Psychology*, 38(4), 311–318. <https://doi.org/10.1037/pap0000372>
- Blöte A. W., Miers A. C., Van den Bos E., & Westenberg P. M. (2019). Negative social self-cognitions: How shyness may lead to social anxiety. *Journal of Applied Developmental Psychology*, 63, 9–15. <https://doi.org/10.1016/j.appdev.2019.05.003>
- Borg, M. E., & Willoughby, T. (2022). Affinity for solitude and motivations for spending time alone among early and mid-adolescents. *Journal of Youth and Adolescence*, 51(1), 156–168. <https://doi.org/10.1007/s10964-021-01520->
- Bowker, J. C., Rubin, K. H., & Coplan, R. J. (2016). Social withdrawal. In R. J. R. Levesque (Ed.), *Encyclopedia of adolescence* (2nd ed., pp. 1–14). Springer.
- Burnell, K., George, M. J., & Underwood, M. (2021). New media and solitude: Implications for peer relations. In R. J. Coplan, J. C. Bowker, & L. J. Nelson (Eds.), *The handbook of solitude: Psychological perspectives on social isolation, social withdrawal, and the experience of being alone* (2nd ed., pp. 254–267). Wiley-Blackwell.
- Campbell, S. W., & Ross, M. Q. (2022). Re-conceptualizing solitude in the digital era: From “being alone” to “noncommunication.” *Communication Theory*, 32(3), 387–406. <https://doi.org/10.1093/ct/qtab021>
- Cheek, J. M., & Buss, H. A. (1981). Shyness and sociability. *Journal of Personality and Social Psychology*, 41, 330–339. <https://doi.org/10.1037/0022-3514.41.2.330>
- Chuk, N. (2018). Media shy: The perils of bashfulness in the digital age, an era of speed, satisfaction, and spectacle. *Interações: Sociedade e as Novas Modernidades*, 34, 92–112. <https://doi.org/10.31211/interacoes.n34.2018.a5>
- Clark, J. L., Algoe, S. B., & Green, M. C. (2018). Social network sites and well-being: The role of social connection. *Current Directions in Psychological Science*, 27(1), 32–37. <https://doi.org/10.1177/0963721417730833>
- Coplan, R. J., Girardi, A., Findlay, L. C., & Frohlick, S. L. (2007). Understanding solitude: Young children’s attitudes and responses towards hypothetical socially-withdrawn peers. *Social Development*, 16, 390–409.
- Coplan, R. J., Hipson, W. E., & Bowker, J. C. (2021). Social withdrawal and loneliness in adolescence: Examining the implications of too much and not enough solitude. *Journal of Youth and Adolescence*, 50(6), 1219–1233. <https://doi.org/10.1007/s10964-020-1365-0>
- Coplan, R. J., Ooi, L. L., & Baldwin, D. (2019). Does it matter when we want to be alone? Exploring developmental timing effects in the implications of unsociability. *New Ideas in Psychology*, 53, 47–57. <https://doi.org/10.1016/j.newideapsych.2018.01.001>
- Currie, C. E., Elton, R. A., Todd, J., & Platt, S. (1997). Indicators of socioeconomic status for adolescents: The WHO health behaviour in school-aged children survey. *Health Education Research*, 12(3), 385–397. <https://doi.org/10.1093/her/12.3.385>
- Daly, O., & Willoughby, T. (2020). A longitudinal person-centered examination of affinity for aloneness among children and adolescents. *Child Development*, 91, 2001–2018.
- Danneel, S., Maes, M., Vanhalst, J., Bijttebier, P., & Goossens, L. (2018). Developmental change in loneliness and attitudes toward aloneness in adolescence. *Journal of Youth and Adolescence*, 47, 148–161. <https://doi.org/10.1007/s10964-017-0685-5>
- Dennis, A. R., Fuller, R. M., & Valacich, J. S. (2008). Media, tasks, and communication processes: A theory of media synchronicity. *MIS Quarterly*, 32(3), 575–600. <https://doi.org/10.2307/25148857>
- Diefenbach, S., & Borrmann, K. (2019). The smartphone as a pacifier and its consequences: Young adults’ smartphone usage in moments of solitude and correlations to self-reflection. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (pp. 1–14). <https://doi.org/10.1145/3290605.3300536>
- Doey, L., Coplan, R. J., & Kingsbury, M. (2014). Bashful boys and coy girls: A review of gender differences in childhood shyness. *Sex Roles*, 70(7–8), 255–266. <https://doi.org/10.1007/s11199-013-0317-9>
- Gardner, W. L., Pickett, C. L., & Knowles, M. (2005). Social snacking and shielding: Using symbols, selves, and surrogates in the service of belonging needs. In K. D. Williams, J. P. Forgas, & W. von Hippel (Eds.), *The social outcast: Ostracism, social exclusion, rejection, and bullying* (pp. 227–242). Taylor & Francis Group.
- Goossens, L. (2014). Affinity for aloneness and preference for solitude in childhood: Linking two research traditions. In R. J. Coplan & J. C. Bowker (Eds.), *The handbook of solitude: Psychological perspectives on social isolation, social withdrawal, and being alone* (pp. 150–166). Wiley-Blackwell.
- Hipson, W. E., Coplan, R. J., Dufour, M., Watanabe, L. K., Wood, K. R., & Bowker, J. C. (2021). Time alone well spent? A person-oriented analysis of adolescents’ solitary activities. *Social Development*, 30(4), 1114–1130. <https://doi.org/10.1111/sode.12518>
- Hollis, C., Livingstone, S., & Sonuga-Barke, E. (2020). Editorial: The role of digital technology in children and young people’s mental health—A triple-edged sword? *Journal of Child Psychology and Psychiatry*, 61, 837–841. <https://doi.org/10.1111/jcpp.13302>
- Keijsers, L., Branje, S. J. T., Frijns, T., Finkenauer, C., & Meeus, W. (2010). Gender differences in keeping secrets from parents in adolescence. *Developmental Psychology*, 46(1), 293–298. <https://doi.org/10.1037/a0018115>

- Keles, B., McCrae, N., & Grealish, A. (2020). A systematic review: The influence of social media on depression, anxiety and psychological distress in adolescents. *International Journal of Adolescence and Youth*, 25(1), 79–93. <https://doi.org/10.1080/02673843.2019.1590851>
- Kock, N. (2004). The psychobiological model: Towards a new theory of computer-mediated communication based on Darwinian evolution. *Organization Science*, 15(3), 327–348. <https://doi.org/10.1287/orsc.1040.0071>
- Kushlev, K., Proulx, J., & Dunn, E. W. (2016). “Silence your phones”: Smartphone notifications increase inattention and hyperactivity symptoms. In *Proceedings of the Emotion Dynamics of Solitude 240. 2016 CHI conference on human factors in computing systems* (pp. 1011–1020). <https://doi.org/10.1145/2858036.2858359>
- Larson, R. W. (1990). The solitary side of life: An examination of the time people spend alone from childhood to old age. *Developmental Review*, 10, 155–183. [https://doi.org/10.1016/0273-2297\(90\)90008-R](https://doi.org/10.1016/0273-2297(90)90008-R)
- Marcoen, A., Goossens, L., & Caes, P. (1987). Loneliness in pre-through late adolescence: Exploring the contributions of a multidimensional approach. *Journal of Youth and Adolescence*, 16, 561–577. <https://doi.org/10.1007/BF021-38821>
- Matias, G. P., Nicolson, N. A., & Freire, T. (2011). Solitude and cortisol: Associations with state and trait affect in daily life. *Biological Psychology*, 86, 314–319. <https://doi.org/10.1016/j.biopsycho.2010.12.011>
- Moroney, S. (2019). Rooms of their own: Child experts, house design, and the rise of the child’s private bedroom. *Journal of Family History*, 44(2), 119–144. <https://doi.org/10.1177/0363199019827235>
- Nguyen, T. T., Ryan, R. M., & Deci, E. L. (2018). Solitude as an approach to affective self-regulation. *Personality and Social Psychology Bulletin*, 44, 92–106. <https://doi.org/10.1177/0146167217733073>
- O’Day, E. B., & Heimberg, R. G. (2021). Social media use, social anxiety, and loneliness: A systematic review. *Computers in Human Behavior Reports*, 3, Article 100070. <https://doi.org/10.1016/j.chbr.2021.100070>
- Ottawa-Carleton District School Board (OCDSB). (2012). *2010-2011 OCDSB Student Survey—Grades 7–12*. <https://ocdsb.ca/common/pages/DisplayFile.aspx?itemId=206817>
- Reid, J. (2012). “My room! Private! Keep out! This means you!”: A brief overview of the emergence of the autonomous teen bedroom in post-World War II America. *The Journal of the History of Childhood and Youth*, 5, 419–443. <https://doi.org/10.1353/hcy.2012.0034>
- Thomas, V., & Azmitia, M. (2019). Motivation matters: Development and validation of the Motivation for Solitude Scale—Short Form (MSS-SF). *Journal of Adolescence*, 70, 33–42. <http://dx.doi.org/10.1016/j.adolescence.2018.11.004>
- Thomas, V., Balzer Carr, B., Azmitia, M., & Whittaker, S. (2021). Alone and online: Understanding the relationships between social media, solitude, and psychological adjustment. *Psychology of Popular Media*, 10(2), 201–211. <https://doi.org/10.1037/ppm0000287>
- Thulin, E., & Vilhelmson, B. (2019). More at home, more alone? Youth, digital media and the everyday use of time and space. *Geoforum*, 100, 41–50.
- Twenge, J. M., & Martin, G. N. (2020). Gender differences in associations between digital media use and psychological well-being: Evidence from three large datasets. *Journal of Adolescence*, 79, 91–102. <https://doi.org/10.1016/j.adolescence.2019.12.018>
- Verduyn, P., Lee, D. S., Park, J., Shablack, H., Orvell, A., Bayer, J., Ybarra, O., Jonides, J., & Kross, E. (2015). Passive Facebook usage undermines affective well-being: Experimental and longitudinal evidence. *Journal of Experimental Psychology: General*, 144(2), 480–488.
- Vogels, E. A. (2019, September 9). *Millennials stand out for their technology use, but older generations also embrace digital life*. Pew Research Center. <https://www.pewresearch.org/fact-tank/2019/09/09/us-generations-technology-use/>
- Wilson, T. D., Reinhard, D. A., Westgate, E. C., Gilbert, D. T., Ellerbeck, N., Hahn, C., . . . Shaked, A. (2014). Just think: The challenges of the disengaged mind. *Science*, 345, 75–77. <http://dx.doi.org/10.1126/science.1250830>
- Wood, K. R., Coplan, R. J., Hipson, W. E., & Bowker, J. C. (2021). Normative beliefs about social withdrawal in adolescence. *Journal of Research on Adolescence*, 32(1), 372–381. <https://doi.org/10.1111/jora.12617>